Simple Linear Regression Analysis

Liam Bengston, Yuze Wang, Jacob Koewler

Choosing the correct numeric variable for SLR

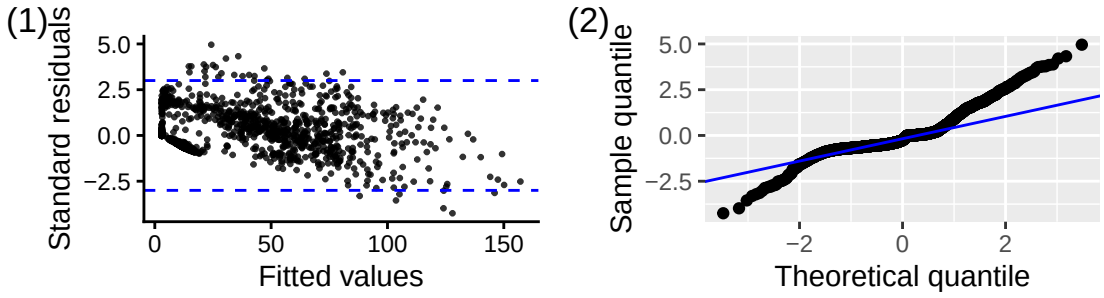
To select a singular variable, we conducted simple linear regression (SLR), we looked at the individual R^2 if we conducted SLR with each numeric variable in the dataset.

Table 1: Top 3 R squared values for numeric predicators in SLR

Predictor variables	R squared values
CO2.tCO2.	0.9733
Lagging_Current_Reactive.Power_kVarh	0.8039
Lagging_Current_Power_Factor	0.1572

From table 1, we see that `CO2.tCO2.` (parts per million emissions of the steel plant), has the highest R^2. However, to better study one of our research questions - how do the electrical properties of the plant affect its total power usage - we dropped `CO2.tCO2` and considered the next best variable for SLR (`Lagging_Current_Reactive.Power_kVarh`) with $R^2 = 0.8039$.

Assess model assumptions

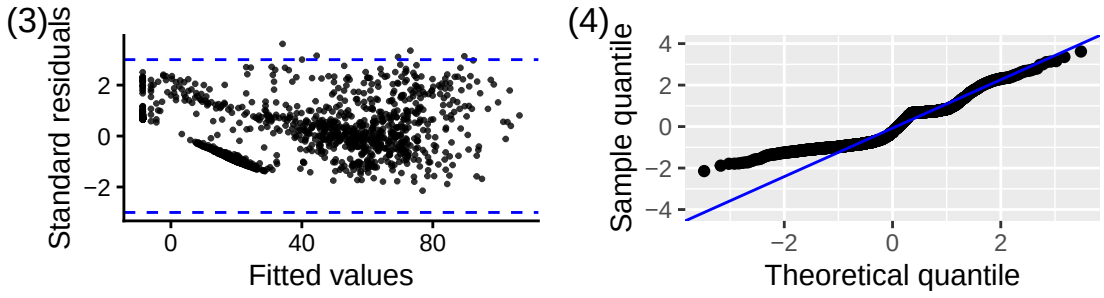


(1) Standard residuals plot and (2) normal QQ plot for y ~ x

Model assumptions for $y \sim x$:

The residual plot (1) shows a clear downward trend, suggesting a violation of the linearity assumption, warranting a transformation, (1) also shows no obvious change in spread, so

the equal variance assumption appears satisfied. Independence cannot be assessed without further information about the factory processes/data collection. The QQ plot (2) shows substantially heavier tails than a normal distribution, violating normality.



(1) Standard residuals plot and (2) normal QQ plot for y ~ sqrt(x)

Model assumptions for $y \sim \sqrt{x}$:

The residual plot (3) shows no obvious pattern, satisfying the linearity assumption. Same statement as the previous model applies to the independence assumption. The QQ plot (4) still shows a heavier left tail, but the improvement suggests that normality is reasonably satisfied. The residual spread in (3) may increase slightly with fitted values, although there is no clear violation, we should proceed with caution.

Model outputs and predictive performance on the test set

Below are model summaries for the finalized SLR model:

```
##             Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) -8.59310  0.6238295 -13.77476 2.622003e-41
## sqrt.LCRP   12.58813  0.1734197  72.58768 0.000000e+00
```

We declare y as the total power usage (kWh) and x as the lagging current reactive power (kVARh). Let the true population model be, with constants β_0, β_1 and iid $\epsilon \sim N(0, \sigma^2)$:

$$y = \beta_0 + \beta_1\sqrt{x} + \epsilon$$

From the summary output, the regression equation is $\hat{y} = -8.593 + 12.588\sqrt{x}$

When the square root of the lagging current reactive power is 0, the estimated power usage is -8.593 kWh. With each unit increase in the square root of the lagging current reactive power, the estimated power usage increases by 12.588 kWh.

The testing set has MSE = 292.152, where MSE is calculated by dividing SSE by $n - 2$ where n is the total number of cases in the testing set. This value stays consistent with the training set (MSE = 300).

List of references, contribution and AI statements

References

Data can be accessed from https://archive.ics.uci.edu/dataset/851/steel+industry+energy+consumption

Contributions

Liam Bengston: Model constructions and selection of relevant models

Jacob Koewler: Model constructions and selection of relevant models

Yuze Wang: Model assumptions and visualizations

AI

Chatgpt was used to help with formatting the latex generating this pdf.